

RANPilotAI

AI-Powered RAN Performance Optimization Platform

Graduation Project ■ 2026

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Agenda

- 1 Introduction, Architecture & KPI Degradation
- 2 KPI Forecasting Engine
- 3 Optimization Action Analyzer
- 4 Telecom RAG Assistant
- 5 Software Testing & Experimental Results

Tasneem Amein

Problem Statement ▪ Proposed Solution ▪ Overall Architecture ▪
KPI Degradation Analyzer

Problem Statement

[Fill in your content here]

Project Motivation & Objectives

[Fill in your content here]

Proposed Solution Overview

[Fill in your content here]

Overall System Architecture

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screenshots/tasneem/hub_landing.png

KPI Degradation Analyzer — Module Overview

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KPI Categories Monitored

[Fill in your content here]

Detection Pipeline & Root Cause Analysis

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KPI Degradation Analyzer — Interface

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KPI Degradation — Sample Results

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Ibrahim Samy

KPI Forecasting Engine ▪ Models ▪ Feature Engineering ▪
Accuracy & Results

Forecasting Module Overview & Motivation

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Forecasting Approach — XGBoost & Holt-Winters

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Feature Engineering & Seasonality Decomposition

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Forecasting Module — Interface

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Model Evaluation & Accuracy Metrics

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7-Day KPI Forecast — Output

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XGBoost Feature Importance

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Khaled Mogahed

Optimization Action Analyzer ▪ Configuration Tracking ▪ Impact Quantification

Action Analyzer — Module Overview

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Configuration Tracking with Dolt

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Action Analyzer — Interface

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Action Timeline View

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Before / After KPI Impact Analysis

Before Configuration Change

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After Configuration Change

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KPI Delta Bar Chart

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Mahmoud Ashraf

Telecom RAG Assistant ▪ Pipeline ▪ Interface ▪
Anti-Hallucination Strategy

RAG Assistant — Overview & Motivation

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RAG Pipeline Architecture

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RAG Assistant — Main Interface

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Live Query — Pipeline Execution

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Retrieved Chunks & Source Citations

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Answer Panel — Citation-Backed Response

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Mohab Tarek

Software Testing ▪ Real-Case Experimental Results ▪ Conclusion
▪ Future Work

Testing Strategy & Test Coverage

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Unit & Integration Testing

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Real-Case Dataset Description

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KPI Degradation — Experimental Results

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Forecasting — Experimental Results

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Action Analyzer — Experimental Results

Configuration Change Event

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KPI Impact Measured

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RAG Assistant — Experimental Results

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Conclusion

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Thank You

Questions & Discussion